where $\boldsymbol{J}$ is the skew symmetric symplectic structure matrix and

$$\tilde{H}_h = H + h^r H_r + h^{r+1} H_{r+1} + \dots \tag{3.10}$$

We have worked out formulas above for the first terms in the expansion in the case of several splitting methods. We can thus view (at least locally) our numerical trajectory as the stroboscopic map (snapshots) of the evolution of a continuous Hamiltonian system.

Ideally, we would like to be able write that the solution to the modified equations is equivalent to mapping via the specified one-step method:

$$\mathscr{G}_h = e^{h\mathcal{L}_{\tilde{f}_h}}.$$

Unfortunately, this statement is elusive. The series expansion in question rarely converges. It is easy to show that for the harmonic oscillator, the series does indeed converge, but it turns out that this is an anomaly. The reason for the lack of convergence of the series is, precisely, its global nature (which is also its chief benefit). In order to proceed, one normally has to truncate the series at some finite number of terms. A natural question is where should we terminate the series? If we include only the $\mathcal{O}(h^r)$ term, then we have not really gained much. For example we would not be able to say that $\tilde{H}_h$ is essentially a conserved quantity along the numerical solutions.

The analysis of the expansion was first treated systematically in [31, 359], where it was shown that a *finite* truncation to terms of order h^k can be achieved in such a way that the error is "exponentially small" in the stepsize. (See also [163, 310].) Define

$$\bar{H}_k \stackrel{\text{def}}{=} H + h^r H_r + h^{r+1} H_{r+1} + \dots + h^k H_k. \tag{3.11}$$

Let us state and prove the following basic result (see [162, 164, 227]):

Theorem 3.1 *Suppose the Hamiltonian H and modified Hamiltonian $\bar{H}_k$ are smooth functions globally defined on a convex, compact subset $\mathcal{B}$ of $\mathbb{R}^{2N_c}$ and suppose that the exact solution and numerical approximations (for h sufficiently small) are confined to $\mathcal{B}$. Then, we have asymptotically for $h \to 0$ that*

$$H(\boldsymbol{z}_n) = H(\boldsymbol{z}_0) + \mathcal{O}(h^r),$$

for $n = 0, 1, \dots, \nu$ where $\tau = \nu h = \mathcal{O}(h^{-k+r})$.

Proof First observe that due to smoothness and the assumptions on $\mathcal{B}$, we have a global Lipschitz constant

$$|H(\boldsymbol{u}) - H(\boldsymbol{v})| \leq L\|\boldsymbol{u} - \boldsymbol{v}\|$$

for all $\boldsymbol{u}, \boldsymbol{v} \in \mathcal{B}$.